

SIGN PATTERNS OF REAL POWERS OF INFINITE PRODUCTS: RESOLUTION OF CONJECTURES 20, 23 AND 24 OF SCHLOSSER AND ZHOU

JAIDEEP SAI PADHI

ABSTRACT. Schlosser and Zhou conjectured precise sign patterns for the coefficients of real powers $P(q)^\delta$ of several infinite products. We settle three of their conjectures: those for the Borwein products $G_7 = (q; q)_\infty / (q^7; q^7)_\infty$ and G_{11} , and for $Q_{12} = (q, q^{11}; q^{12})_\infty / (q^5, q^7; q^{12})_\infty$.

- The conjecture for G_{11} is true.
- The conjecture for G_7 fails exactly for $\delta_c < \delta < 5$, where $\delta_c = 4.873507585386734263\dots$ is a root of the 897-th coefficient polynomial. It holds at $\delta = 5$ and on all other stated ranges.
- The conjecture for Q_{12} holds for $\delta = \pm 1$ and for $2 \leq \delta \leq 3$. In the range $-1 < \delta < 0$ it holds exactly for $-1 < \delta \leq \delta_1$, where $\delta_1 = -0.644113480813\dots$ is a root of the 22nd coefficient polynomial.

In each failure a secondary term of the circle-method expansion, carrying the wrong sign, overtakes a leading amplitude that vanishes nearby. The proofs combine explicit Rademacher expansions, with all constants certified in ball arithmetic, and exact polynomial computations for small n . They use the framework of a companion paper on the Göllnitz–Gordon product.

1. INTRODUCTION

For a product $P(q) = \prod_{m \geq 1} (1 - q^m)^{\epsilon(m)}$ and $\delta \in \mathbb{R}$ write $P(q)^\delta = \exp(\delta \log P) = \sum_n c_\delta(n) q^n$, where $\log P(0) = 0$. Taking logarithmic derivatives gives

$$n c_\delta(n) = \delta \sum_{k \leq n} b_k c_\delta(n - k), \quad b_k = - \sum_{d|k} \epsilon(d) d.$$

In particular each $c_\delta(n)$ is a polynomial in δ with rational coefficients.

Schlosser and Zhou [5, Appendix A] conjectured the sign patterns listed below. As in [5], a sign pattern allows zero coefficients.

- **Conjecture 20** ($P = G_7$). The pattern is $+-00+0$ for $\delta = 1$; $+-+-++-$ for $2 \leq \delta < 3$; $+ - 0 + 00 -$ for $\delta = 3$; and $+-+-+--$ for $3 < \delta \leq 5$.
- **Conjecture 23** ($P = G_{11}$). The pattern is $+-0-+0+000$ for $\delta = 1$; $+-+-++-+-+$ for $\gamma \leq \delta \leq 2$; and $+ - 0 + - 0 - 000 +$ for $\delta = 3$. Here $\gamma = 1.7584535519419110333\dots$ is the root in $(1.5, 2)$ of $c_\delta(21)$.
- **Conjecture 24** ($P = Q_{12}$). The pattern is $+-+0-+-+ - 0 + -$ for $\delta = 1$; $+-+-+ - + - + - + -$ for $2 \leq \delta \leq 3$; $+++++0 - - - - - 0$ for $\delta = -1$; and $+++++ - - - - - +$ for $-1 < \delta < 0$.

The printed strings in [5] are partly illegible in electronic form. The patterns above are the ones determined by the coefficients: they agree with every coefficient we computed, and the forced zeros come from classical identities (§2).

Theorem 1.1. *Let $\delta_c = 4.87350758538673426336\dots$ be the unique root of $c_\delta(897)$ (for G_7) in an interval of width $1.3 \cdot 10^{-19}$ around this value. Conjecture 20 holds for $\delta = 1$, for $2 \leq \delta \leq 3$, for $3 < \delta \leq \delta_c$, and for $\delta = 5$. It fails for every $\delta \in (\delta_c, 5)$.*

Date: September 10, 2026.

Theorem 1.2. *Conjecture 23 holds.*

Theorem 1.3. *Conjecture 24 holds for $\delta = \pm 1$ and for $2 \leq \delta \leq 3$. For $-1 < \delta < 0$ the pattern $++++- - - - - +$ holds if and only if $\delta \leq \delta_1$, where $\delta_1 = -0.644113480813476518842928\dots$ is the unique root of $c_\delta(22)$ in $(-1, 0)$. For instance, $c_{-1/4}(6) = 545/65536 > 0$ even though the pattern requires a minus sign at $n = 6$.*

Together with the companion paper [4], which treats Conjecture 21 (the Göllnitz–Gordon product Q_8), and earlier work of He–Li [3] and Bringmann–Heim–Kane [2], every conjecture of [5, Appendix A] is now resolved: Conjectures 17–19 and 22 were settled in [3, 2]. For integer exponents with $m(t-1) \leq 24$, complete sign characterizations of G_t^m were obtained by Wang [6]; this covers $\delta \in \{1, 2, 3, 4\}$ for G_7 . Our certificates for these integer cases are independent of [6].

The mechanism of failure. For G_7 at residue $n \equiv 1 \pmod{7}$ three terms compete:

- the leading term, at the cusps of denominator 7. Its amplitude $A_7(1, \delta)$ vanishes at $\delta^* = 4.8735075853867550889\dots$
- the leading term at the cusps of denominator 14. Its amplitude is ± 0.0195 , with the wrong sign for $n \equiv 1 \pmod{14}$.
- the correction term, which grows for $\delta > 4$ and has the correct sign.

Just below δ^* the wrong-sign term wins on a window of n bounded on both sides, and the window first becomes nonempty at $n = 897$. The true threshold δ_c is therefore a root of $c_\delta(897)$, lying $2.1 \cdot 10^{-14}$ below δ^* . This is the phenomenon behind the threshold $8/3$ in [4], but it is realized at finite n . For Q_{12} with $-1 < \delta < 0$ the failure is a plain small-coefficient phenomenon: $c_\delta(22)$ changes sign at δ_1 and remains of the wrong sign on $(\delta_1, 0)$.

2. FORCED ZEROS AND EXACT VANISHING

- The zeros of G_7^δ at $\delta = 1$ (residues 3, 4, 6) and of G_{11} at $\delta = 1$ (residues 3, 6, 8, 9, 10) follow from Euler’s pentagonal number theorem: $m(3m-1)/2$ avoids these residues, and $(q^p; q^p)_\infty$ is a series in q^p .
- The zeros at $\delta = 3$ (residues 2, 4, 5 for G_7 , and 2, 5, 7, 8, 9 for G_{11}) follow in the same way from Jacobi’s identity $(q; q)_\infty^3 = \sum (-1)^m (2m+1) q^{m(m+1)/2}$; see also [5, Cor. 13].
- For $Q_{12}^{\pm 1}$ the zeros at $n \equiv 3 \pmod{6}$, respectively $n \equiv 5 \pmod{6}$, are instances of the Andrews–Bressoud vanishing theorem [1].

3. TRANSFORMATION FORMULAE AND CUSP STRUCTURE

3.1. Borwein products. We use the transformation of [5, Prop. 7], which follows from the classical eta transformation. There is one correction: the auxiliary integer must satisfy $hh' \equiv -1 \pmod{k}$, the Hardy–Ramanujan convention, not $hh' \equiv 1$.

Let $\tau = h/k + iz/k^2$ and $d = \gcd(p, k)$. Then

$$G_p(e^{2\pi i \tau})^\delta = \left(\frac{p}{d}\right)^{\delta/2} e^{\pi i \delta (-s(h,k) + s(ph/d, k/d))} \exp\left(\frac{\pi \delta (d^2 - p)}{12pz} - \frac{(p-1)\pi \delta z}{12k^2}\right) \widehat{G}^\delta,$$

where

$$\widehat{G} = \frac{f(e^{2\pi i d h'_d/k - 2\pi d^2/(pz)})}{f(e^{2\pi i h'/k - 2\pi/z})}, \quad f(x) = \prod_{m \geq 1} (1 - x^m)^{-1},$$

and all powers are principal.

Lemma 3.1 (certified branch). *Consider every cusp h/k with $k \in \{7, 14, 21, 28\}$ for $p = 7$, or $k \in \{11, 22\}$ for $p = 11$. For each, the difference between $\log G_p(\tau)$, computed from its q -series, and the logarithm of the right-hand side, built from principal logarithms and the series for $\log f$, is $2\pi i \cdot 0$ on the connected set $\{\operatorname{Re} z > 0, \operatorname{Re}(1/z) \geq 1\}$.*

Proof. Both sides are continuous on this set, and their difference lies in $2\pi i\mathbb{Z}$, so it is constant. At $z = 1$ it was evaluated in 200-bit ball arithmetic, with explicit tail bounds, and has modulus at most 10^{-19} . With the convention $hh' \equiv +1$ the same difference is about $5.8 \cdot 10^{-4}$, which is not an integer; this confirms the corrected convention. $\square$

When $p \mid k$ we have $\widehat{G} = 1 - e(-h'/k)e^{-2\pi/z} + \dots$, with $hh' \equiv 1$ in this display. The main term grows like $e^{\pi\delta(p-1)/(12z)}$, and the correction term like $e^{\pi(\delta(p-1)/12-2)/z}$. The correction term therefore grows exactly when $\delta > 24/(p-1)$; for $p = 7$ this means $\delta > 4$. When $p \nmid k$ the cusp is bounded:

$$|G_p^\delta| \leq K_p^\delta, \quad K_p = \sqrt{p} e^{-\pi(p-1)/(12p)} f(e^{-2\pi/p}) f(e^{-2\pi}),$$

so that $K_7 = 4.8195$ and $K_{11} = 13.734$.

3.2. The theta quotient Q_{12} . Put $\vartheta_c(\tau) = \sum_{J \equiv c(48)} e(\tau J^2/96)$. Then

$$Q_{12} = q^{-1} \frac{\vartheta_{10} - \vartheta_{14}}{\vartheta_2 - \vartheta_{22}}.$$

As in [4], the projective orbit of the pair of numerator and denominator under the Weil representation at level 96 was computed exactly in $\mathbb{Z}[\zeta_{192}]$, modulo $\Phi_{192} = x^{64} - x^{32} + 1$.

- The orbit has 48 elements and 10 cusp types.
- Exactly one type grows for $\delta > 0$. Its leading exponents are $(J_1, J_5) = (2, 10)$, its support is $J \equiv \pm 2, \pm 10 \pmod{24}$, and all its coefficients have equal modulus. Exactly one type grows for $\delta < 0$, namely the inverse type.
- An elementary Gauss-sum argument, as in [4, Lemma 3.5], shows that growing cusps have $4 \mid k$. An exact check of all $k \leq 36$ gives: h/k grows iff $12 \mid k$ and $h \equiv 5, 7 \pmod{12}$ for $\delta > 0$, and $h \equiv 1, 11 \pmod{12}$ for $\delta < 0$.
- The suprema of $|R^{\pm 1}|$ over the bounded types on $\operatorname{Im} \tau' \geq 1$ are at most $K_{12} = 550.75$ and $K'_{12} = 547.23$. These were certified by branch-and-bound.
- Exact Gauss sums and certified branch integers at the cusps of denominators 12 and 24 give the amplitudes below.

For $\delta > 0$ the growth is $e^{2\pi\delta/z}$ and the amplitudes are

$$2 \cos \frac{5\pi n}{6} \text{ (main)}, \quad -2\delta \cos \frac{\pi(5n-1)}{6} \text{ (first correction)}, \quad \delta(\delta+1) \cos \frac{\pi(5n-2)}{6} \text{ (second correction)}.$$

For $\delta = -s < 0$ the amplitude at $k = 12$ is $2 \cos \frac{\pi(n-2s)}{6}$, and at $k = 24$ it is $2 \cos \frac{\pi(n-2s)}{12} + 2 \cos \frac{\pi(11n+2s)}{12}$.

4. EXPLICIT EXPANSIONS AND ERROR CONSTANTS

In every case the Rademacher dissection is carried out as in [4, §4]: Ford-circle arcs, chords of length at most $2\sqrt{2}k/(N+1)$, and $N \geq \sqrt{4\pi n} + 1$. The main terms have the shape $\mathcal{P}_k(B) = \frac{2\pi}{k} \sqrt{B/C} I_1(\frac{2\pi}{k} \sqrt{BC})$.

- For G_p : $B = \delta(p-1)/12$ for the main term, $B - 2$ for the correction term, and $C = 2n - \delta(p-1)/12$.
- For Q_{12} : $B = 2\delta, 2(\delta-1), 2(\delta-2)$ for the main term and the two corrections, and $C = 2(n+\delta)$.

The error is bounded by $\mathcal{E} = \mathcal{E}_{\text{arc}} + \mathcal{E}_{\text{rem}} + \mathcal{E}_{\text{ng}}$. For G_p these are

$$\mathcal{E}_{\text{arc}} = \sqrt{2\pi} \frac{p-1}{p^2} e^{1+\pi\delta(p-1)/12} (1+\delta e^{-2\pi}[\cdot]), \quad \mathcal{E}_{\text{rem}} = 2\sqrt{2} \frac{p-1}{p^2} e^{1+\pi\delta(p-1)/12} (f(t)^\delta f(t^p)^\delta - 1 - \delta t[\cdot]), \quad \mathcal{E}_{\text{ng}} = 2\sqrt{2}$$

with $t = e^{-2\pi}$, and where $[\cdot]$ means the term is included only when the correction is extracted, i.e. $\delta > 24/(p-1)$. For Q_{12} the analogues come from the majorant of the growing type. Typical values are $\mathcal{E} \approx 1.9 \cdot 10^4$ for G_7 at $\delta = 4.87$, and $\mathcal{E} \approx 1.6 \cdot 10^3$ for G_{11} at $\delta = 2$.

Lemma 4.1 (envelopes). *For $x > 0$, $\kappa(x_0)U(x) \leq I_1(x) \leq U(x)$ whenever $x \geq x_0$, where $U(x) = e^x/\sqrt{2\pi x}$ and $\kappa(x_0) = \sqrt{2\pi x_0} e^{-x_0} I_1(x_0)$ [4, Lemma 5.2]. Consequently, if every competitor term is bounded through U and the dominant term from below through κU , then each ratio competitor/dominant is at most $Cy^a e^{-gy}$ with $y = \sqrt{C}$ and a gap $g > 0$. Such a bound is decreasing for $y \geq a/g$. Hence a single verification at $n = n_1$ proves the inequality for every $n \geq n_1$.*

All *dominance* certificates below are of this form. In each, the gaps satisfy $g \geq 0.1$, the exponents satisfy $a \leq 3$, and $y(n_1) \geq 30$.

5. PROOF OF THEOREM 1.1

Throughout this section $p = 7$. The main-term amplitude at a cusp of denominator k ($7 \mid k$) is

$$a_k(n, \delta) = \sum_h \cos(-\pi\delta\sigma_h - 2\pi hn/k), \quad \sigma_h = s(h, k) - s(h, k/7),$$

and the correction amplitude is

$$b_k(n, \delta) = -\delta \sum_h \cos(-\pi\delta\sigma_h - 2\pi hn/k - 2\pi h'/k), \quad hh' \equiv 1 \pmod{k}.$$

Both sums run over the $\varphi(k) \leq 6k/7$ residues h coprime to k . With $B = \delta/2$, $B_2 = \delta/2 - 2$ and $C = 2n - \delta/2$, Lemma 4.1 and the expansion of §4 give

$$(1) \quad c_\delta(n) = \sum_{7 \mid k \leq N} \left(a_k(n, \delta) \mathcal{P}_k(B) + [\delta > 4] b_k(n, \delta) \mathcal{P}_k(B_2) \right) + E, \quad |E| \leq \mathcal{E}(\delta).$$

5.1. Auxiliary bounds.

Lemma 5.1. *Let $7 \mid k$ and let s lie in a compact subinterval of $(0, 5]$.*

- (a) $|\sigma_h| < k/12 + k/84 \leq k/10.5$. Hence $|\partial_s(\pi s \sigma_h)| \leq \pi k/10$, and $|\partial_s a_k| \leq \pi k \varphi(k)/10$.
- (b) On every Farey chord of order N , $|\text{Im} \log G_7| \leq 0.556N^2$. Moreover $|\log |G_7|| \leq \log K_7 + 0.449N^2 + 2$ at non-growing cusps.
- (c) (arc lemma) Let $A_s = \int_{\text{arc}} \Phi(s) \exp(\pi s/(2z) + \pi(2n - s/2)z/k^2) dz$, taken over either small arc of K from 0. Then

$$|\partial_s A_s| \leq e^{1+\pi s/2} \left[\frac{\pi k}{10} \cdot \frac{\pi}{\sqrt{2}} \cdot \frac{k}{N+1} + \frac{2+\pi/2}{s/2} + \frac{\pi^2}{\sqrt{2}k(N+1)} \right].$$

The same bound holds for the correction integrand, with $s/2$ replaced by $s/2 - 2 > 0$ in the middle term and an extra factor $se^{-2\pi}$.

- (d) (Bessel derivative) With $x = \frac{2\pi}{k} \sqrt{BC}$,

$$|\partial_s \log \mathcal{P}_k| \leq D_k := \frac{1}{4B} + \frac{1}{4C} + \left(1 + \frac{1}{x}\right) \frac{2\pi}{k} \frac{\sqrt{C}}{4\sqrt{B}}.$$

Proof. (a) For $k \geq 2$ the classical bound $|s(h, k)| \leq (k-1)(k-2)/(12k)$ applies.

(b) Write $\log G_7 = \sum_{7|m} \log(1 - q^m)$ with principal logarithms, and use $|\operatorname{Arg}(1 - x)| \leq \frac{\pi}{2}|x|$ for $|x| < 1$. This gives $|\operatorname{Im} \log G_7| \leq \frac{\pi}{2}(1 - |q|)^{-1}$. On a chord $\operatorname{Re} z \geq k^2/(2N^2)$, so $1 - |q| \geq 0.9\pi/N^2$. For $\log |G_7|$, use the transformation formula at $d = 1$ together with $|\widehat{G}| \in [(f(e^{-2\pi w/7})f(e^{-2\pi w}))^{-1}, f(e^{-2\pi w/7})f(e^{-2\pi w})]$, where $w = \operatorname{Re}(1/z) \leq 2N^2/k^2$.

(c) With $w = 1/z = 1 + iv$ one has $\partial_s f_s = (i\theta + \pi w/2 - \pi/(2k^2 w))f_s$, where $|\theta| \leq \pi k/10$ by (a). The terms $i\theta f_s$ and $\pi f_s/(2k^2 w)$ are bounded by the arc length times $e^{1+\pi s/2}$. For $\frac{\pi}{2} \int w f_s dz = -\frac{\pi}{2} \Phi i e^{\pi s/2} \int e^{i\pi s v/2} g(v) dv$, with $g = e^{b/(1+iv)}/(1+iv)$, integrate by parts exactly as in [4, Lemma 7.2]; this gives the factor $(2 + \pi/2)/(s/2)$.

(d) Differentiate $\log \mathcal{P}_k$ in s , using $I'_1/I_1 \leq 1 + 1/x$. The latter follows from $I_0 = I_2 + 2I_1/x \leq I_1 + 2I_1/x$. $\square$

Summing (c) over the growing cusps ($7 \mid k \leq N$, at most $6k/7$ of them for each k) introduces the harmonic factor $\sum_{7 \mid k \leq N} 1/k \leq \frac{1}{7}(1 + \log(N/7))$. The same summation over the chords, with lengths at most $2\sqrt{2}k/(N+1)$, bounds the derivative of the remaining error pieces:

- on chords at growing cusps, by $(|\theta| + \frac{\pi}{2}|1/z| + 0.1)(F_s - 1 - [\cdot]st) + F_s \ell_7 - [\cdot]t$, where $F_s = f(t)^s f(t^7)^s$, $\ell_7 = \log(f(t)f(t^7))$ and $|1/z| \leq 2N^2/k^2$;
- on chords at non-growing cusps, by K_7^s times the bound of (b).

Together this gives an explicit quantity $\mathcal{E}'(s, N) = O(N^3 \log N)$, which is implemented in `g7_deriv.py` and `g7_eps3.py`.

5.2. Steps of the proof. *Step 1 (exact small- n computations).* The exact polynomials $c_\delta(n) \in \mathbb{Q}[\delta]$ were computed for $n \leq 897$. Descartes' rule of signs, applied after the Möbius map of the interval to $(0, \infty)$ and with bisection, shows the following.

- On $[3, \delta_{\text{lo}}]$ every $c_\delta(n)$ has sign $\sigma(n \bmod 7)$ and no root. Where $c_3(n) = 0$, the factor $(\delta - 3)$ is first divided out exactly.
- On $[\delta_{\text{lo}}, \delta_{\text{hi}}]$ only $c_\delta(897)$ has a root, and it has exactly one. This defines δ_c .
- On $[2, 3]$ every $c_\delta(n)$ has the pattern $+-+++-$. Zeros at $\delta = 3$ are divided out, and the sign of the factor $(\delta - 3) \leq 0$ is taken into account.

Step 2 (dominance, $n \geq 898$). Let $J = [a, b]$ be a δ -interval, fix a residue r , and set $n_1 \geq 898$ with $n_1 \equiv r \pmod{7}$. The amplitude $a_7(r, \cdot)$ is enclosed on J in centred form, $a_7(r, m) + a'_7(r, J)(J - m)$, which is exact for linear functions. Suppose

$$\min_J |a_7(r, \delta)| \cdot \frac{2\pi}{7} \sqrt{\frac{a/2}{C_{\max}}} \kappa(x_1) U(x_1) > [b > 4] 6b \mathcal{P}_7(B_2) + \sum_{k=14,21} \varphi(k) (\mathcal{P}_k(B) + b \mathcal{P}_k(B_2)) + \frac{N}{7} \cdot \frac{6N}{7} (1+b) \mathcal{P}_{28}(B) +$$

where $x_1 = \frac{2\pi}{7} \sqrt{(a/2)C_{\min}}$, $C_{\min/\max} = 2n_1 - b/2$ resp. $2n_1 - a/2$, and $B = b/2$ on the right. where every $\mathcal{P}$ on the right is replaced by its upper envelope. Then $\operatorname{sgn} c_\delta(n) = \operatorname{sgn} a_7(r, \delta)$ for all $\delta \in J$ and all $n \geq n_1$ with $n \equiv r \pmod{7}$, by Lemma 4.1. Adaptive bisection certifies this on

$$[2, 3 - 10^{-9}], \quad [3 + 10^{-9}, 4.87], \quad [4.87, 4.87350758538]$$

with $n_1 = 898$, and on the window $W = [4.87350758538, \delta_c]$ for all residues $r \neq 1$ with $n_1 = 898$, and for $r = 1$ with $n_1 = 2500$.

Step 3 (near $\delta = 3$).

Proposition 5.2. *Let $r \in \{2, 4, 5\}$ and $0 < |\delta - 3| \leq 10^{-9}$. Then for every $n \geq 898$ with $n \equiv r \pmod{7}$, $\operatorname{sgn} c_\delta(n) = \operatorname{sgn} ((\delta - 3) a'_7(r, 3))$, which is the conjectured sign on each side of 3.*

Proof. By §2, $c_3(n) = 0$, and $a_7(r, 3) = 0$ exactly; the latter was checked in $\mathbb{Z}[\zeta_{28}]$. Subtract (1) at δ and at 3, with the same N . The $k = 7$ term becomes $a_7(r, \delta)\mathcal{P}_7(\delta/2)$, and

$$|a_7(r, \delta)| \geq |\delta - 3| \left(|a'_7(r, 3)| - |\delta - 3| \sum_h (\pi\sigma_h)^2 \right).$$

Every other difference is at most $|\delta - 3|$ times a bound from Lemma 5.1: for $k \geq 14$ this is $\varphi(k)(\pi k/10 + D_k)\mathcal{P}_k$, and for the error it is $\mathcal{E}'(s, N)$. Dividing by $|\delta - 3|$ gives a sufficient inequality that no longer involves $\delta - 3$. In envelope form it is certified at $n_1 = 898$ with margin at least $6.7 \cdot 10^8$ on both sides of 3, and by Lemma 4.1 it then holds for all $n \geq 898$. $\square$

The residues 0, 1, 3, 6 with $|\delta - 3| \leq 10^{-9}$ are covered by Step 2.

Step 4 (the window, residue 1, $898 \leq n < 2500$).

Proposition 5.3. *For every $n \geq 898$ with $n \equiv 1 \pmod{7}$ and every $\delta \in [4.87350758538, \delta_{\text{hi}}]$, we have $\partial_\delta c_\delta(n) > 0$.*

Proof. Differentiate (1). The dominant contribution is $a'_7(1, \delta)\mathcal{P}_7(\delta/2)$, where $a'_7(1, \delta) > 0.15$ on the interval. Against it:

- the term $|a_7(1, \delta)| |\partial_s \mathcal{P}_7|$, with $|a_7(1, \delta)| \leq 10^{-11}$ on the interval; its ratio to the dominant term is bounded by its value at $n = 2500$;
- the correction at $k = 7$, whose amplitude derivative is at most $6 + 6\delta\pi \cdot \frac{5}{14}$;
- the terms with $k \geq 14$;
- $\mathcal{E}'$.

The resulting inequality is certified in envelope form at $n_1 = 898$ with margin $3.6 \cdot 10^{10}$. $\square$

The values $c_{\delta_{\text{hi}}}(n)$ are certified negative for all $n \leq 7000$ with $n \equiv 1 \pmod{7}$, $n \neq 897$; this used a 3000-bit ball recurrence. For $898 \leq n < 2500$, $n \equiv 1 \pmod{7}$, Proposition 5.3 then gives $c_\delta(n) < c_{\delta_{\text{hi}}}(n) < 0$ for all $\delta \in [4.87350758538, \delta_c]$. For $n \geq 2500$, Step 2 applies.

Step 5 ($\delta = 1, 3, 5$). This is Step 2 at a single point with $n_1 = 400$, together with exact integer coefficients for $n \leq 400$. At $\delta = 5$, $a_k(r, 5) = 0$ for $k = 7, 14, 21$ and $r \equiv 1, 2, 6 \pmod{7}$, exactly. There the dominant term is $b_7(r, 5)\mathcal{P}_7(1/2)$, whose Bessel argument $\frac{2\pi}{7}\sqrt{C/2}$ exceeds that of every remaining term.

Step 6 (failure on $(\delta_c, 5)$).

- For $\delta \in (\delta_c, \delta_{\text{hi}}]$, $c_\delta(897) > 0$, because δ_c is its only root in $[\delta_{\text{lo}}, \delta_{\text{hi}}]$.
- On $[\delta_{\text{hi}}, 4.87350758538675509]$, an interval containing δ^* , the exact polynomial $c_\delta(897)$ is positive at both endpoints and has no sign variation (Descartes count 0). So it is positive throughout.
- For $\delta \in (\delta^*, 5)$, $a_7(1, \delta) > 0$. By (1) and Lemma 4.1, $c_\delta(n) > 0$ for all sufficiently large $n \equiv 1 \pmod{7}$.

In every case the pattern requires a minus sign at such n . For instance, $c_{49/10}(57) > 0$ by exact rational arithmetic. $\square$

6. PROOF OF THEOREM 1.2

On $[\gamma, 2]$ the $k = 11$ amplitude has no zero, and the smallest amplitude occurs at residue 10 near γ . Dominance holds for all $n \geq 300$ on $[1.7584, 2]$. For $n \leq 310$ the exact polynomials have the pattern on $[\gamma, 2]$. For this check $c_\delta(21)$ is divided by its degree-18 irreducible factor, which vanishes at γ and has no other root in $[\gamma, 2]$, and the zeros at $\delta = 2$ are divided out exactly. For $\delta = 1, 3$ we use dominance for $n \geq 600$ and exact integers for $n \leq 600$; the forced zeros are those of §2. $\square$

7. PROOF OF THEOREM 1.3

$2 \leq \delta \leq 3$. The main amplitude $2 \cos(5\pi n/6)$ vanishes at $n \equiv 3, 9 \pmod{12}$. There the first correction, with amplitude $\mp \delta$, decides the sign. Its Bessel argument exceeds that of the $k = 24$ main term whenever $\delta > 4/3$, so no threshold phenomenon occurs. The criterion is verified explicitly for every $400 \leq n \leq 20000$ on 8 subintervals of δ ; the tail $n > 20000$ is handled on 40 subintervals; and exact polynomials cover $n \leq 400$.

$\delta = \pm 1$. Dominance for $n \geq 181$, exact integer coefficients for $n \leq 400$, and forced zeros by [1].

$-1 < \delta \leq \delta_1$. This is the argument of [4, §7] with the following changes.

- The growing cusps are $h \equiv 1, 11 \pmod{12}$ with $12 \mid k$.
- The main-term Bessel data are $B = 2s$, $C = 2(n-s)$, and the amplitude is $2 \cos(\pi(n-2s)/6)$. Near $s = 1$ this equals $\mp 2 \sin(\pi\varepsilon/3)$ at $n \equiv 5, 11 \pmod{12}$.
- The phase bound is $|\theta_{h,k}| \leq k^2/2 + 0.01$, the minor-arc constant is K'_{12} , and the arc lemma carries the frequency $2\pi s$, which contributes the term $(2 + \pi/2)/s_0$.
- The chord bound is $(k^2/2 + 4\pi N^2/k^2 + 0.2)(\tilde{F}_1 - 1) + \tilde{F}_1 \ell$, and the non-growing derivative bound is $2\sqrt{2}eK'_{12}(\log K_{12} + 0.02 + 0.556N^2)$.

Every sum over k is evaluated explicitly for each n (`q12_neg.py`), so no harmonic factor is lost. At $\delta = -1$ the coefficients with $n \equiv 5, 11 \pmod{12}$ vanish identically. The ε -difference criterion of [4, §7] is uniform in $\varepsilon = \delta + 1 \in (0, 0.356]$. It is verified for $774 \leq n \leq 20000$, and the tail is handled analytically. Exact polynomials on $[-1, \delta_1^+]$ cover $n \leq 780$: the only root in the interval is that of $c_\delta(22)$ at δ_1 . Since $c_\delta(22)$ has the wrong sign on all of $(\delta_1, 0)$, the pattern fails there. $\square$

8. COMPUTATIONS

All certified steps use either exact arithmetic (FLINT) or ball arithmetic (Arb), via `python-flint` 0.9.0. A table of scripts is included in the accompanying repository.

- *Heaviest runs.* The exact root isolations for G_7 with $n \leq 897$ took about 20 minutes per interval, and the computation of the degree-897 polynomials took 3 minutes.
- *Independent check.* As a check on the Descartes computations for G_7 on $[2, 3]$, a separate complex-root isolation was run for $n \leq 160$. It agrees.
- *Mathematical inputs.* The only inputs beyond those of [4] are the classical eta transformation (with the corrected convention, certified in Lemma 3.1), Euler's and Jacobi's identities, and [1].

REFERENCES

- [1] G. E. Andrews and D. M. Bressoud, *Vanishing coefficients in infinite product expansions*, J. Austral. Math. Soc. Ser. A **27** (1979), 199–202.
- [2] K. Bringmann, B. Heim and B. Kane, *On a sign-change conjecture of Schlosser and Zhou*, J. Math. Anal. Appl. **548** (2025).
- [3] B. He and L. Li, *Some conjectures of Schlosser and Zhou on sign patterns of the coefficients of infinite products*, arXiv:2509.10023.
- [4] J. S. Padhi, *Sign patterns of real powers of the Göllnitz–Gordon product: a complete resolution of a conjecture of Schlosser and Zhou*, preprint.
- [5] M. J. Schlosser and N. H. Zhou, *On the infinite Borwein product raised to a positive real power*, Ramanujan J. **61** (2023), 515–543.
- [6] L. Wang, *Sign changes of coefficients of powers of the infinite Borwein product*, Adv. in Appl. Math. **141** (2022), 102405.

DEPARTMENT OF COMPUTER SCIENCE, PURDUE UNIVERSITY, WEST LAFAYETTE, IN, USA
Email address: `jpadhi@purdue.edu`